Assignment 2

SYSC4906G

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Introduction

This assignment outlines a fish schooling system represented as a Cell-DEVS model. The original work is by Mohammad Etemad and implements the model in CD++. This assignment will deal with the conversion, and partial refactoring/redesign of the original work into Cadmium.

Background

In nature, fish form what are called “schools”: a grouping of fish that swim together in a synchronized manner, often with the same speed and direction. Under normal circumstances, fish decide to form a school to reduce the risk associated with their predators, as a large grouping of fish is likely to scare or confuse the predator. Fish, and their school, predict their next move based on several factors including their neighboring fish and the proximity of a predator [1]. In a paper covering fish school formations by Hubbard et al., they observed that members of a school maintain a constant speed, yet often remain in the same position for an extended period of time [1].

Original Model & Modifications

The following will detail the CD++ model created by Mohammad Etemad in the original work. This section of the report will give a brief overview of the original authors work first, before detailing the modifications necessary to convert it to Cadmium.

Conceptual Overview

We will begin with a brief overview of the conceptual model from the original work. The model of the original work is rather simple and primarily focuses on a *vision-based* schooling model. In this model, fish begin to form schools when they see a neighbouring fish and are not already part of a school. It is important to note that the model is purely vision based, and the work does not consider external factors. Some of these factors that are ignored are hot and cold boundary areas and/or predators.

At a very high level, the fish are spawned into an initial configuration from which they are free to move randomly. When a fish that is not already a part of a school sees a neighbouring fish, it will join that fish and its school should one exist already, or the two (or more) fish will form a new school.

CD++ Implementation

The CD++ implementation of this model consists of a three-layer system. We will discuss each layer below.

The first layer represents the *real* environment where the fish are swimming. This is where the simulation begins as the fish are spawned into this initial layer. In this layer, fish are represented by red cells.

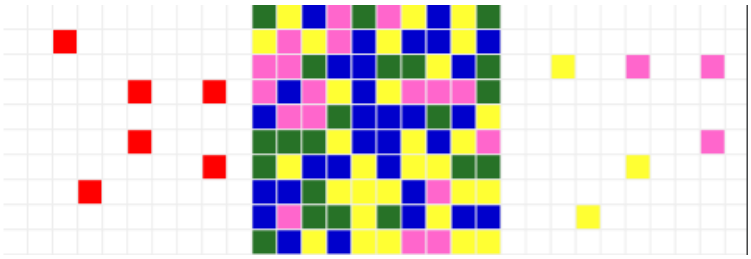
The second layer represents the randomly generated direction each fish will travel in during the next time step.

The third and final layer represents the orientation of each fish. This represents the direction of travel from the previous time step.

Each of the layers is notated by an integer scale, and a colour coated scheme. Integer values are mapped to the four directions a fish can move: East, North, West, and South map to {1, 2, 3, 4}, respectively. From here, a colour scheme is used to represent either their *next* move or the *last* movement a fish took, which is used to represent their orientation. The system is as follows:

- Moving East is represented by GREEN
- Moving North is represented by YELLOW
- Moving West is represented by BLUE
- Moving South is represented by PINK

Here is an example figure of the three layers:



There are a few key things to note from the above. First, we need to look at the rightmost grid (layer 3) first. This layer indicates the *last* movement of each fish. Here we see that three fish moved South, and three moved North. Given this layer represents the last move of each fish, we can look at the leftmost grid (layer 1) to see the current positioning of each fish. In this layer we see that each fish has moved in their respective direction that is found in layer 3. This is also used to represent the orientation of a fish. For example, a fish that just moved South is now facing South.

The final thing to note about the above figure is the second layer represents the random direction assigned to each cell in the *next* time step. This means that for each fish in the first layer, they will move in the randomly assigned direction found in their corresponding cell of layer 2. For example, the fish in the top left-hand corner has a yellow cell in layer 2, thus it will move North its next move.

The final thing to note about the original model is the field of view. The model operates with a Von Neumann neighbourhood scheme, where each fish has a field of view of a singular cell, except for the cell *behind* its current orientation. For example, if a fish is traveling North, it can see the cell in West and East, but not the Southern cell.

Cadmium Modifications

There are two major modifications that will be made to convert this model to Cadmium. The first is that the original three-layer system will be compressed into a single layer, represented as a state struct. This will allow us to capture all the relevant information found in the original three-layers in a single layer (struct).

The second modification is changing to a two-cell vision field instead of one. Although the original work specifies that the fish have a vision field of one cell, the actual implementation extends this field of view to manage the surrounding environment. As such, we will expand the fish vision to two cells. This allows a fish to see another fish that is separated by a singular empty cell in its immediate vicinity. This is visualized in the below table, where the red cell represents the fish and the blue cells represent the fish's field of view. It is important to note that this represents the *potential* field of view and does not consider the orientation. Like in the original CD++ model, when considering orientation the fish cannot see directly behind it. For example, if the fish if facing North it cannot see South.

Conceptual Model

The largest deviation from the original CD++ model is that instead of having a three-layer model, we will compress the model into a singular layer. To do so, each cell will have defined state variables that map to the original three layers. In the model, each cell with have a two-cell viewing radius with a Von Neumann neighborhood scheme.

The cell space will remain 10x10, and the initial positioning of the fish will be randomly generated.

As for rules, we will keep the original rule set. The set is as follows.

The movement of fish can be categorized into two categories:

1. Random movement
2. Movement towards another fish

As with the original work, it is assumed that all fish will move randomly unless another fish is within their viewing radius. If this occurs, then the two (or more) fish will join or form a new school. The specific rules to follow this behaviour are defined in the context of a singular fish and what their surroundings look like. This set is defined as:

**** The original ruleset was defined in terms of up, down, left, right. For consistency, it has been converted to North, South, West, East ****

1. If there is a fish on the West, and you are not in a school, then move West and form a school.
2. If there is a school on the East, and you are not in a school, them move East and join the school.
3. If there is no fish to the East or West, and there is a fish to the North, and you are not in a school, then move North and form a school.
4. If there is no fish to the East or West, and there is a school to the South, and you are not in a school, then move South and form a school.

We also need to define some collision priority. It is as follows:

1. Priority if given to fish moving horizontally.
2. Moving West has higher priority than moving East.
3. Moving North has more priority than moving South.
4. The fish that is *not* given priority will remain in its original cell, well the fish *given* priority will move to the designated cell.

Formal Specification

The formal specification for the atomic Cell-DEVS model is defined as follows:

$$< X, Y, S, N, d, \tau, \delta_{int}, \delta_{ext}, \lambda, D >$$

- $S = \{p, d, o\}$

Where,

  - Presence (p) indicates if the cell is with or without a fish. $p \epsilon \{0, 5\}$, where 0 indicates an empty cell, and 5 indicates the presence of a fish.
  - Direction (d) indicates the next direction the fish will attempt to move. $d \epsilon \{1,2,3,4\}$, mapping to East, North, West, South.
  - Orientation (o) indicates the direction the fish was heading in the previous time step. $o \epsilon \{0,1,2,3,4\}$, where 0 indicates an no movement. The remaining values map the same as d.
- $X = \{x | x \epsilon S\}$
- $Y = \{y | y \epsilon S\}$
- $N = S^5 \rightarrow \{(1,0), (0,1), (0,0), (-1,0), (0,-1)\}\ for\ Von\ Neumann\ Neighborhoods$
- $D = 1\ time\ unit$
- $\tau: N \rightarrow S$

Where the rules are evaluated in the following priority order:

1. **Schooling Anchor:** If a fish is in a school it is anchored to its current cell.
2. **Grouping:** If a fish is isolated with another fish in its viewing radius, it will override its randomly generated direction and move towards the other fish/school.

3. **Random Movement:** If a fish is isolated with no other fish/schools in its viewing radius, it will move in its randomly assigned direction.

- $\delta_{int}, \delta_{ext}, \lambda: All\ defined\ as\ per\ Cell - DEVS\ specifications$

Simulation Results

This section will detail the same experimental framework as the original report. We will test the following scenarios:

1. Vertical fish alignment
2. Horizontal fish alignment
3. Cross formation
4. Right angle formation
5. Diamond formation
6. Distributed Initialization

For each of the above simulations, we will include a screenshot of the key time stamps. Additionally, the Cell DEVS Viewer has been configured to show, in order of left to right, the *presence, direction,* and *orientation* values of the associated state.

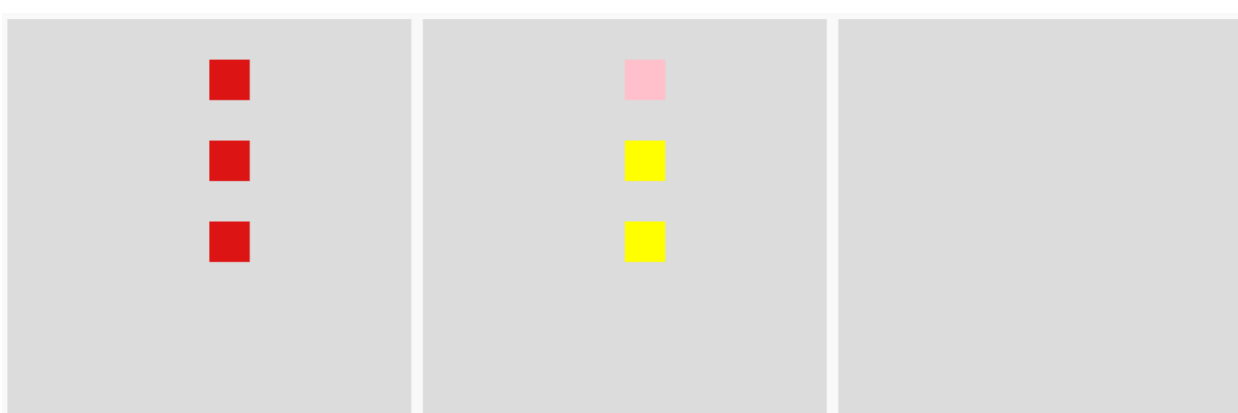
Also note that some of the random directions are different than that of the original work. This does not impact the test framework, however it should not be treated as an *exact* one-to-one of the original work's simulations.

The full log files and videos of execution can be found in the repository.

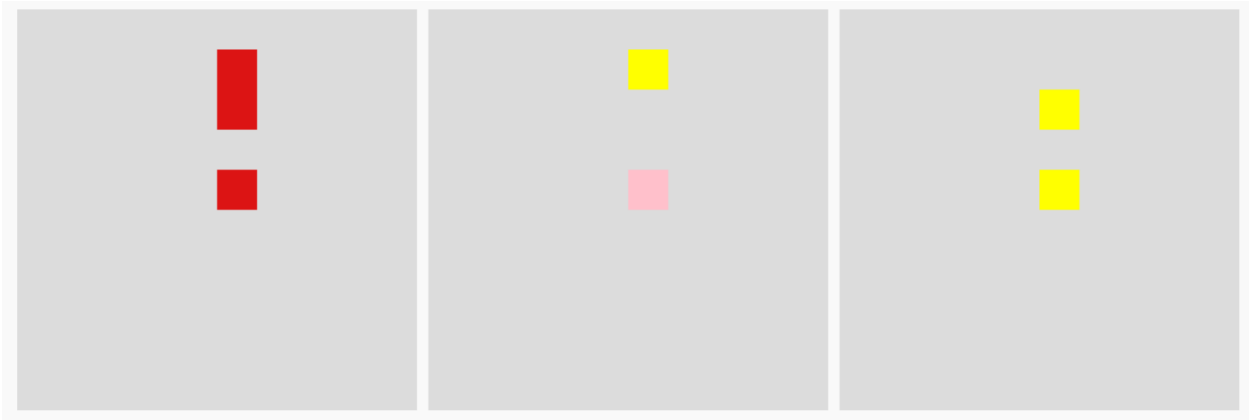
Vertical Alignment

The following scenario shows how the fish vertically attract.

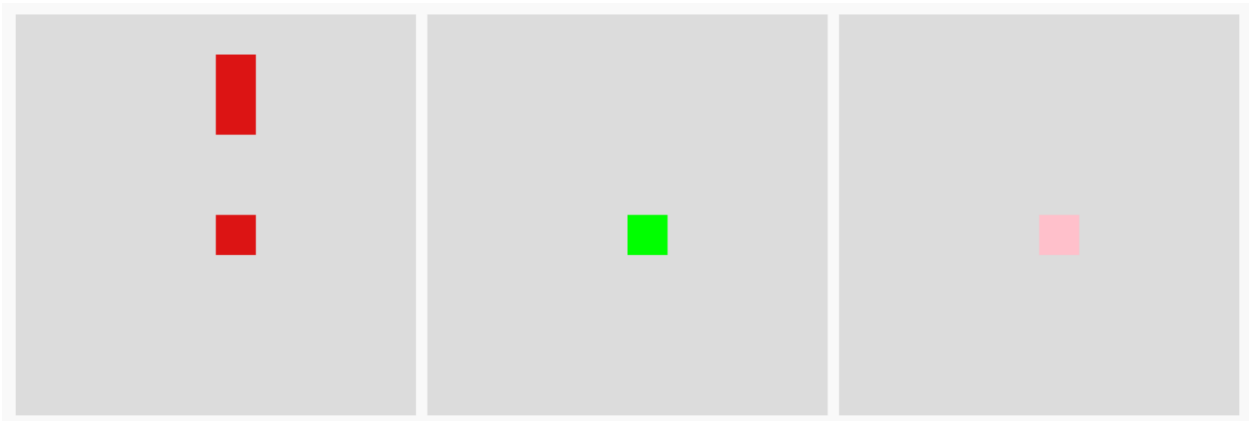
Initial Configuration:



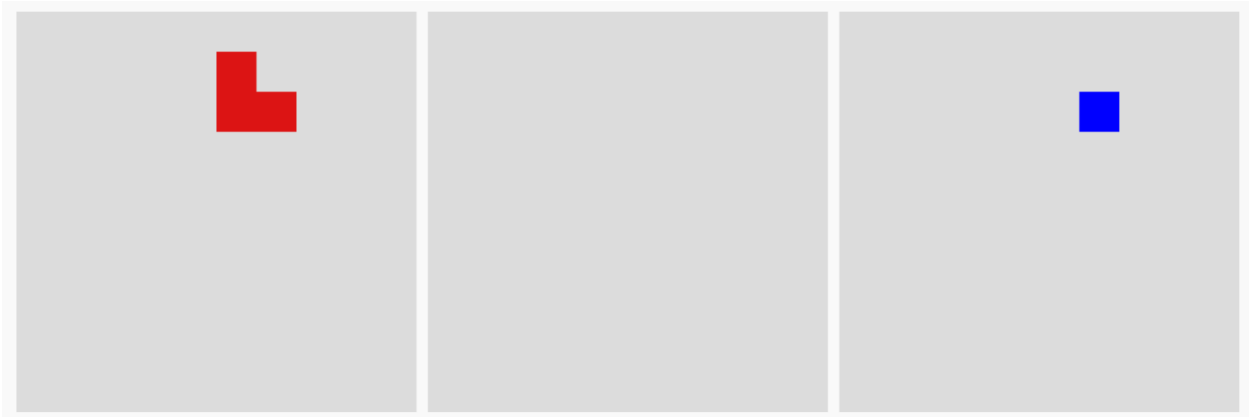
At t = 1:



At t=2:



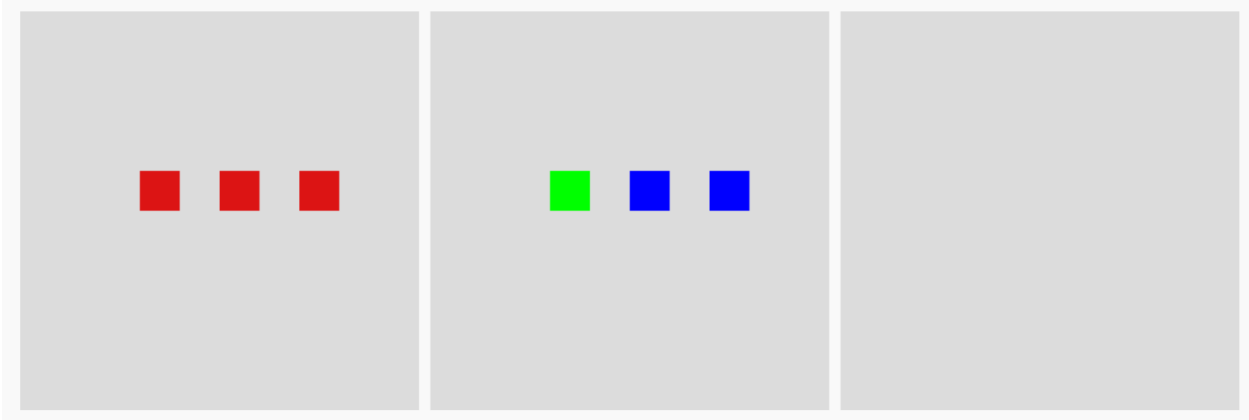
Final configuration: t=8



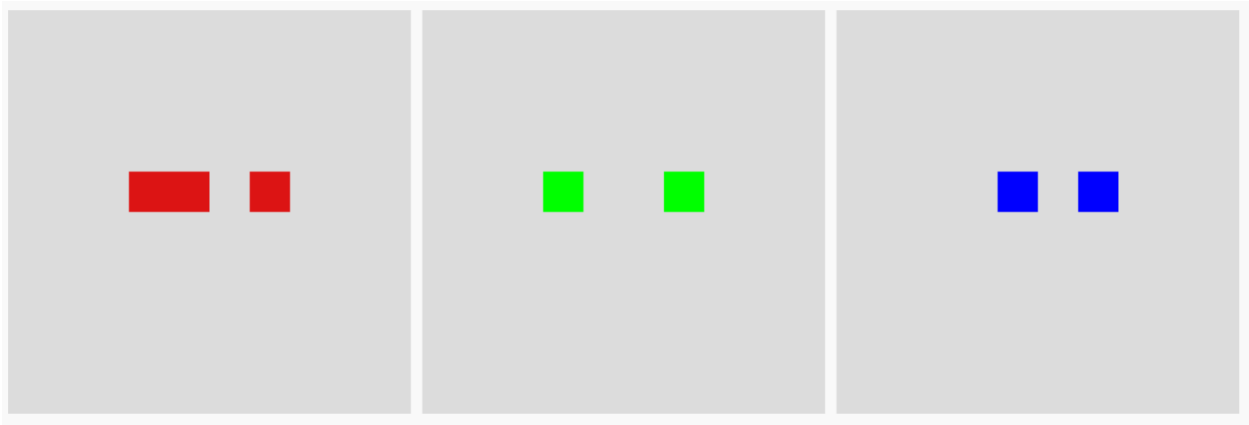
Horizontal Alignment

The following scenario shows how the fish horizontally attract.

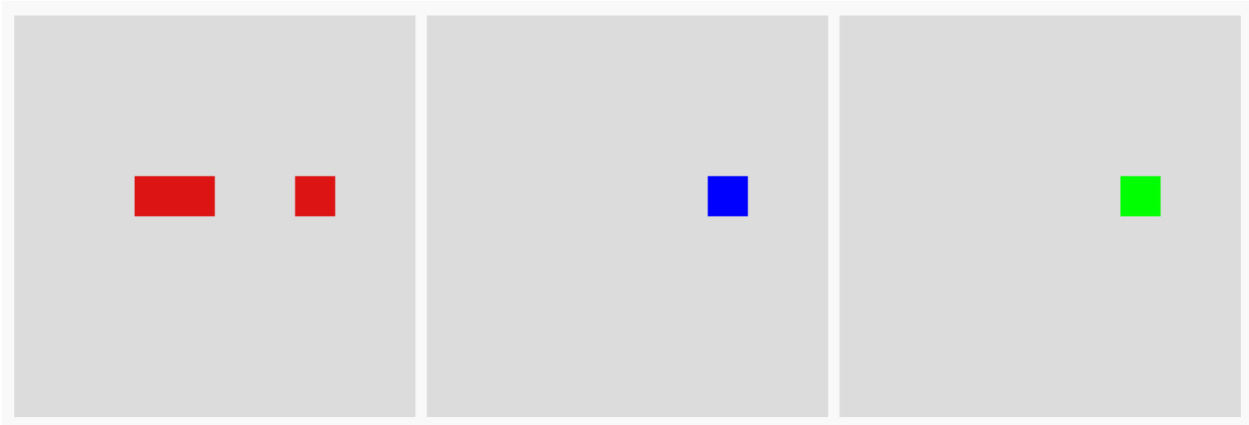
Initial Configuration:



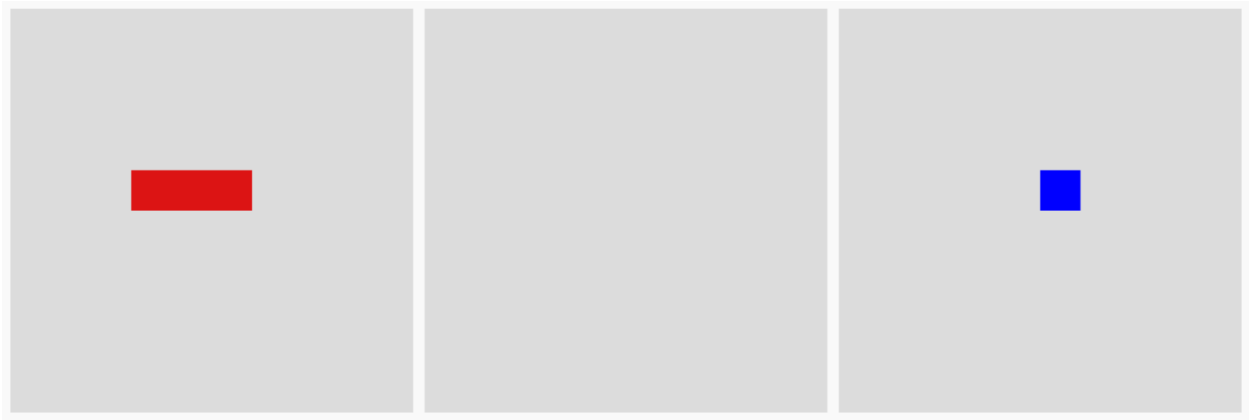
T=1



T=2

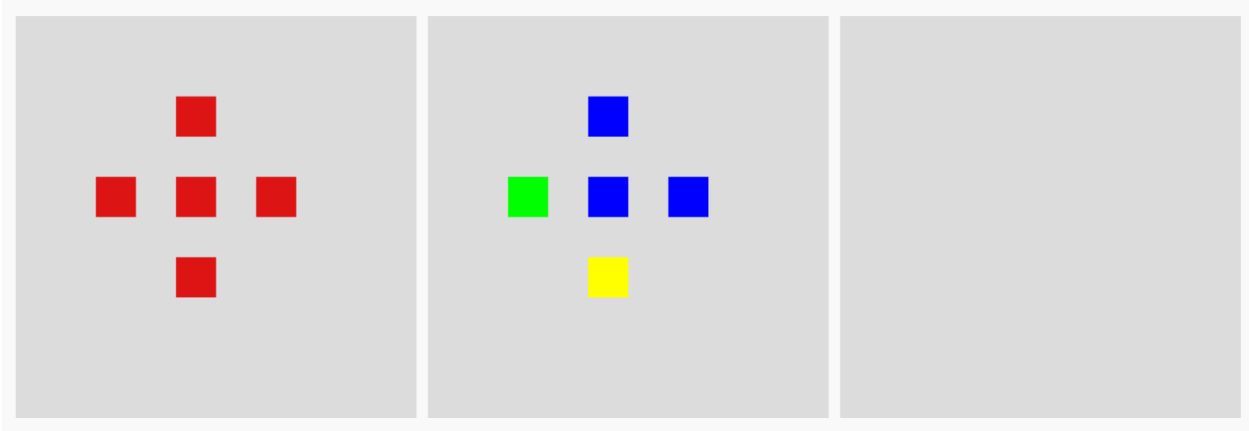


T=4

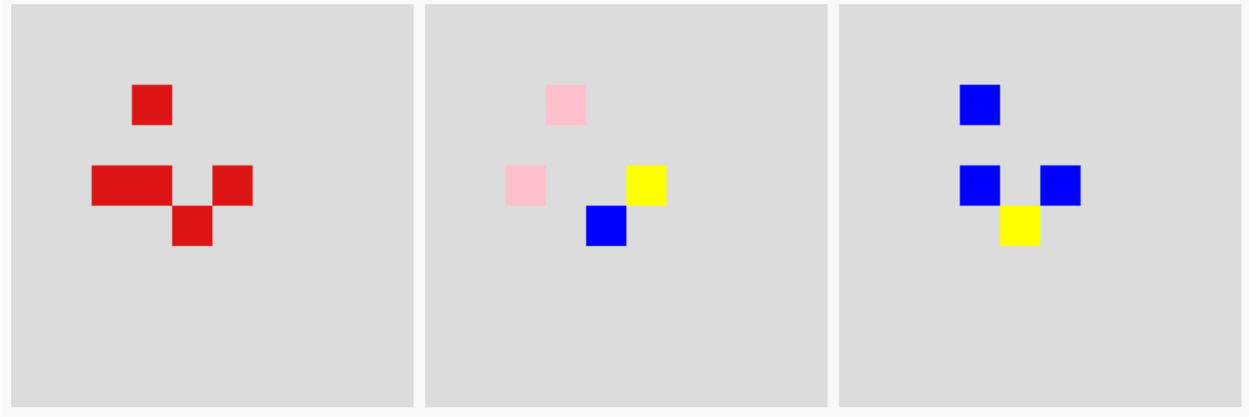


Cross Formation

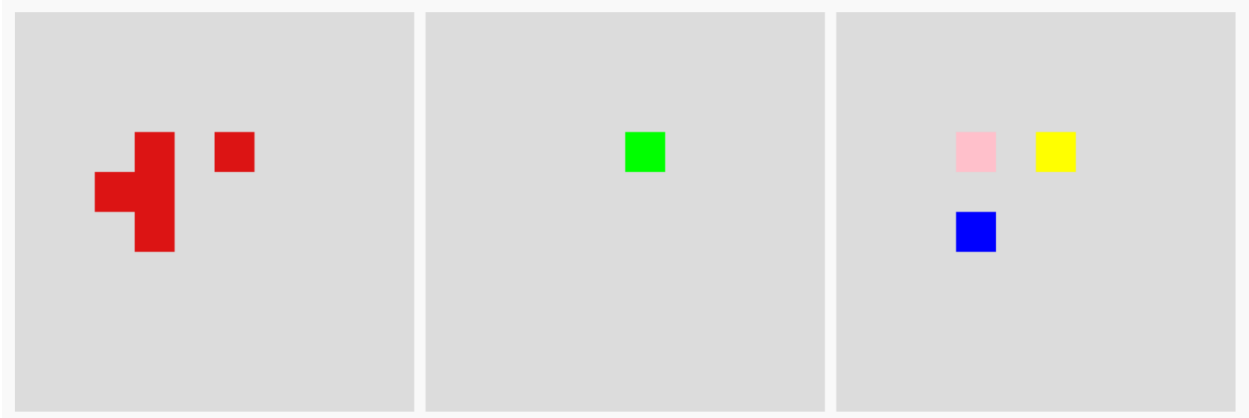
Initial Configuration:



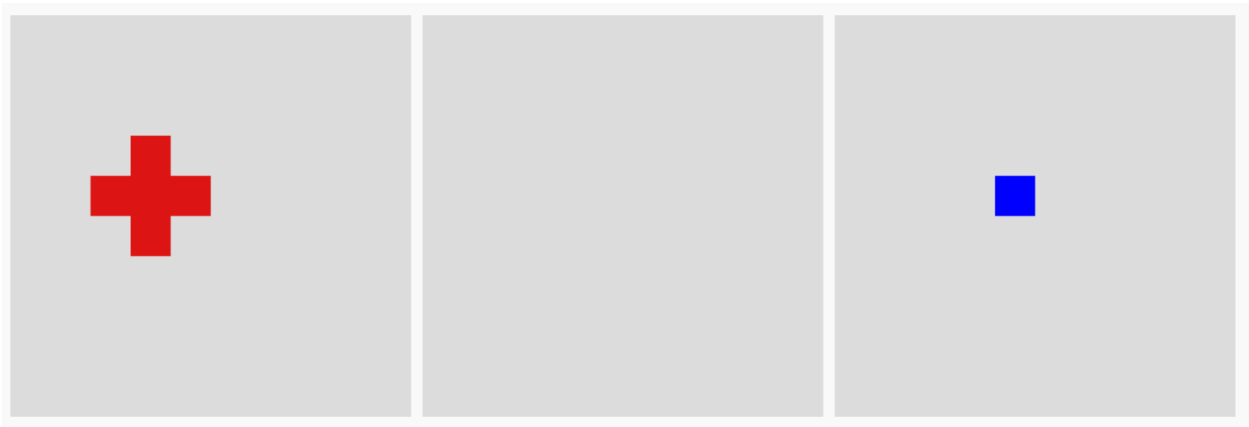
T=1



T=2



Final Configuration (t=6)

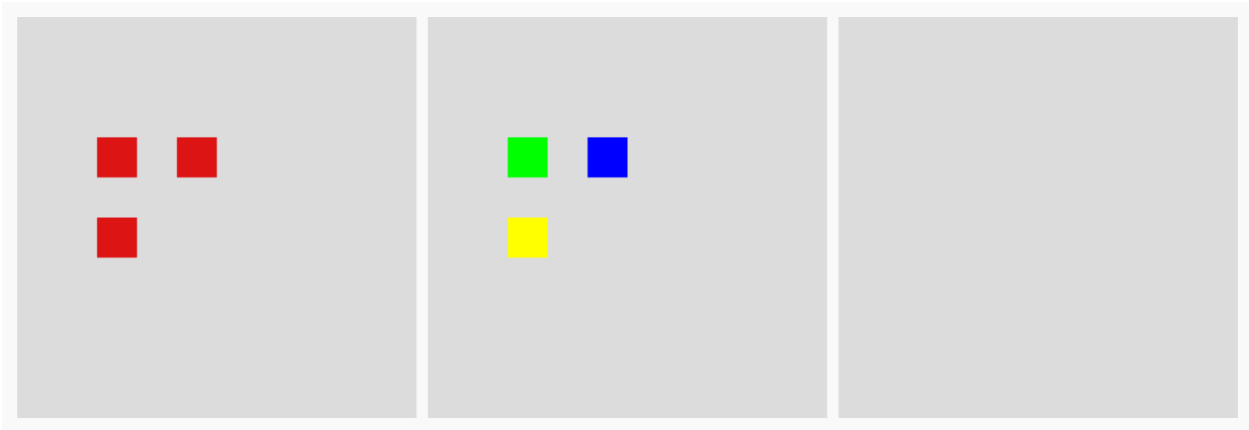


Right Angle Formation

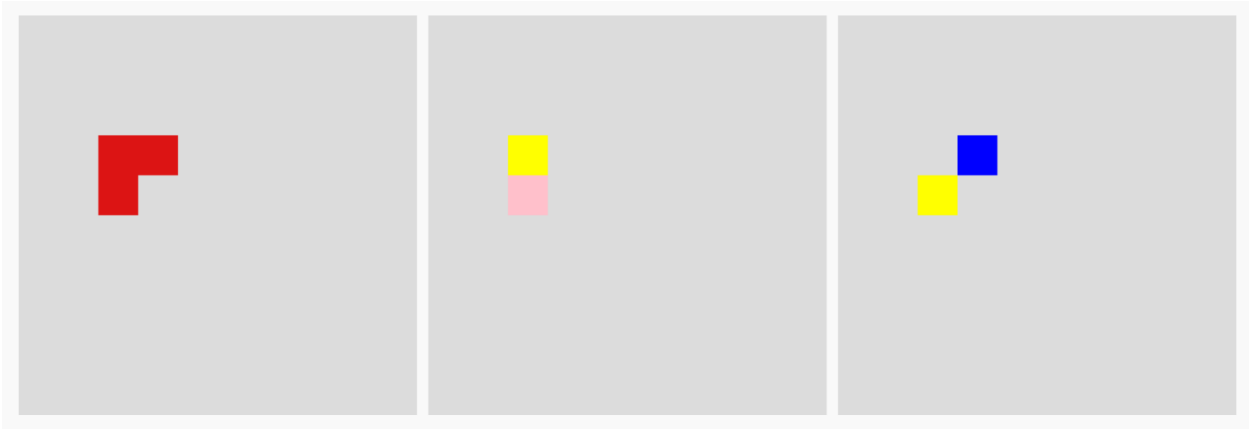
This scenario demonstrates how a fish anchors to form a school with one or more fish.

Initial Configuration:

T=1



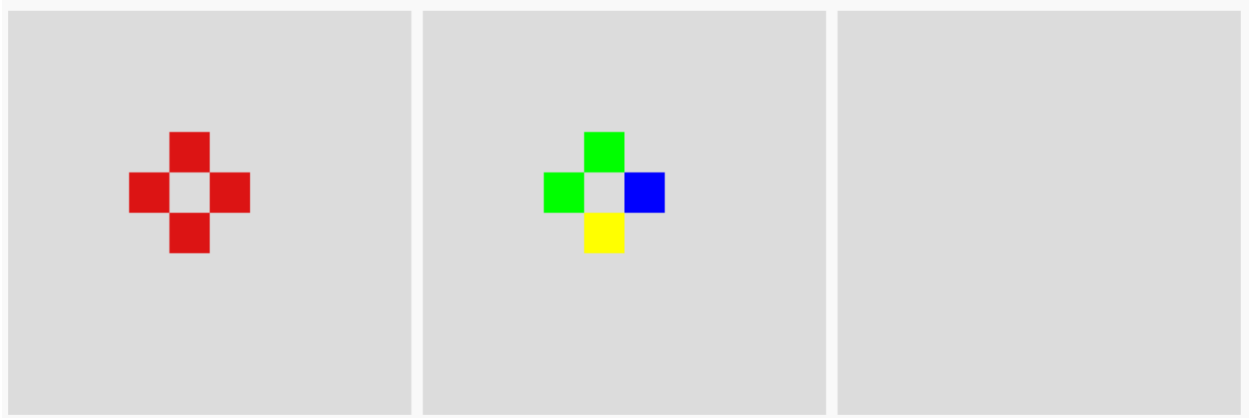
T=2



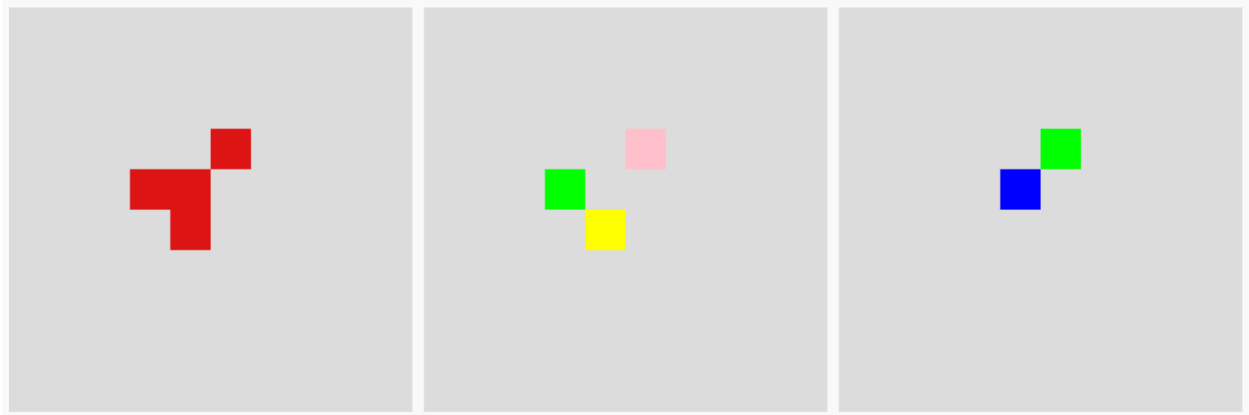
Diamond Formation

This scenario demonstrates how the priority movement rules work.

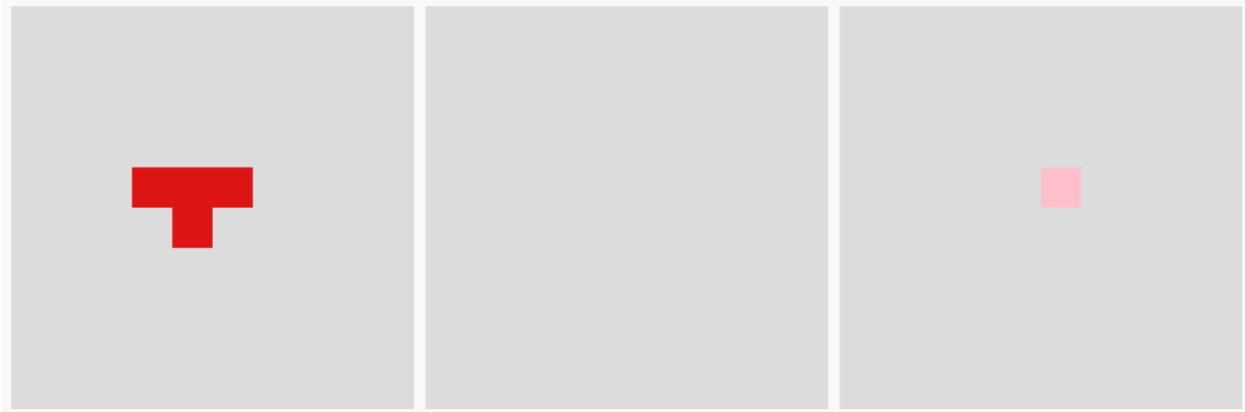
Initial Configuration:



T=1



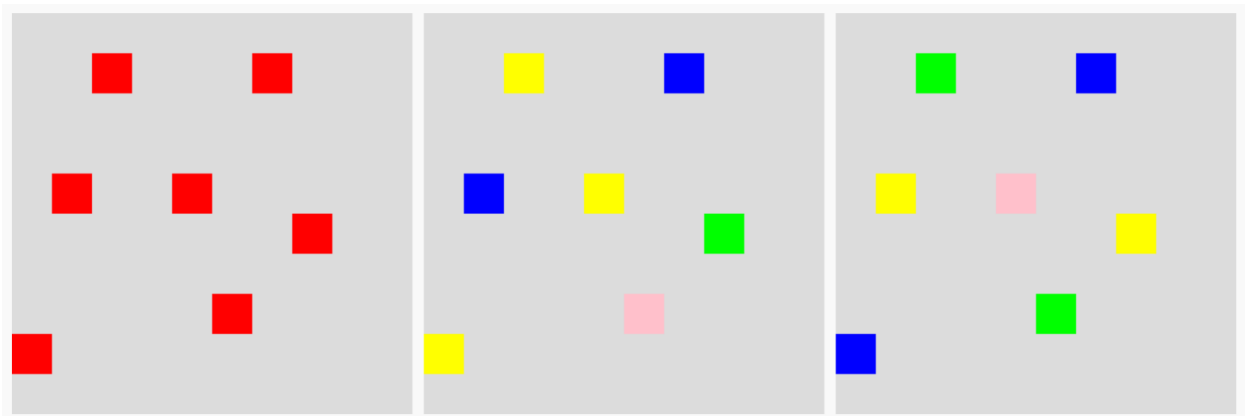
Final configuration (T=2)



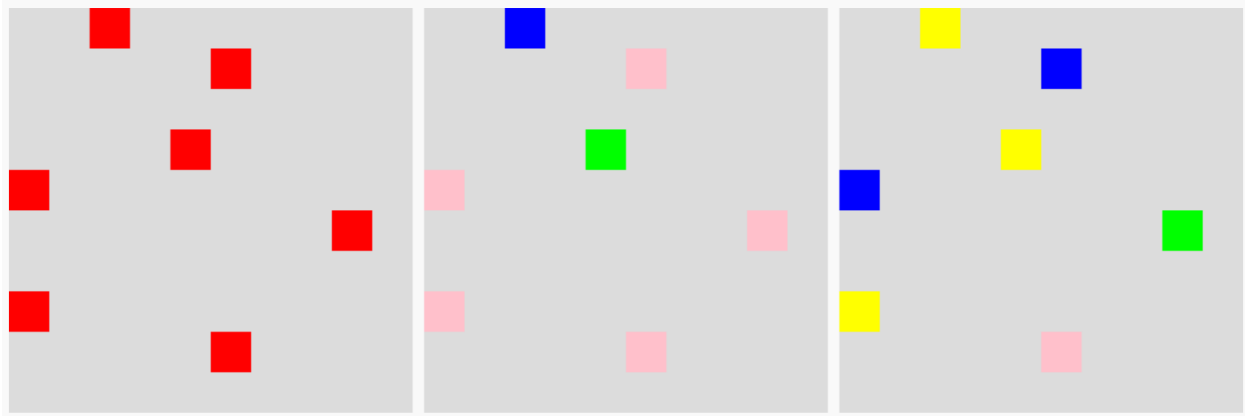
Distributed Initial Configuration

This scenario shows a more natural flow of the system, where fish move in random directions and interact with their neighbors according to the implemented rule set.

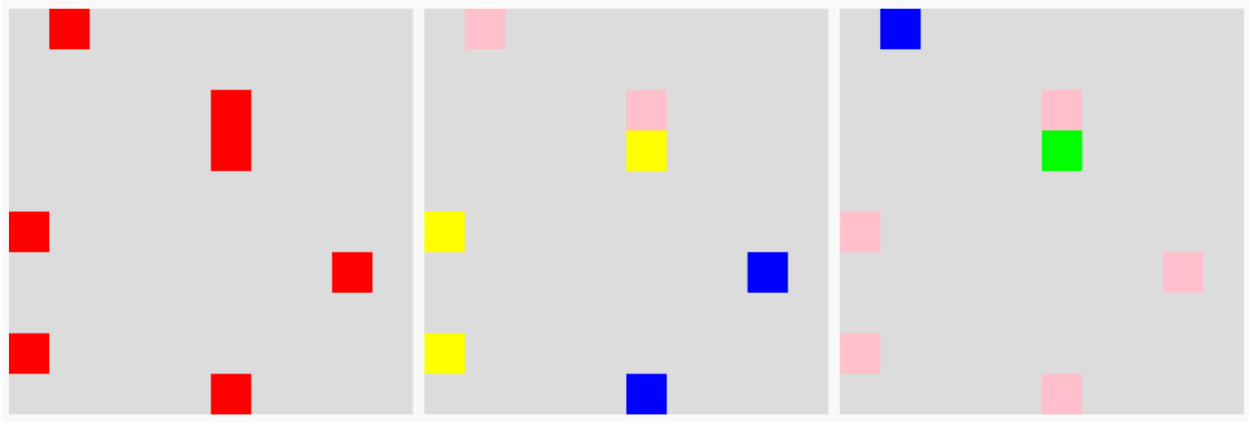
Initial Configuration:



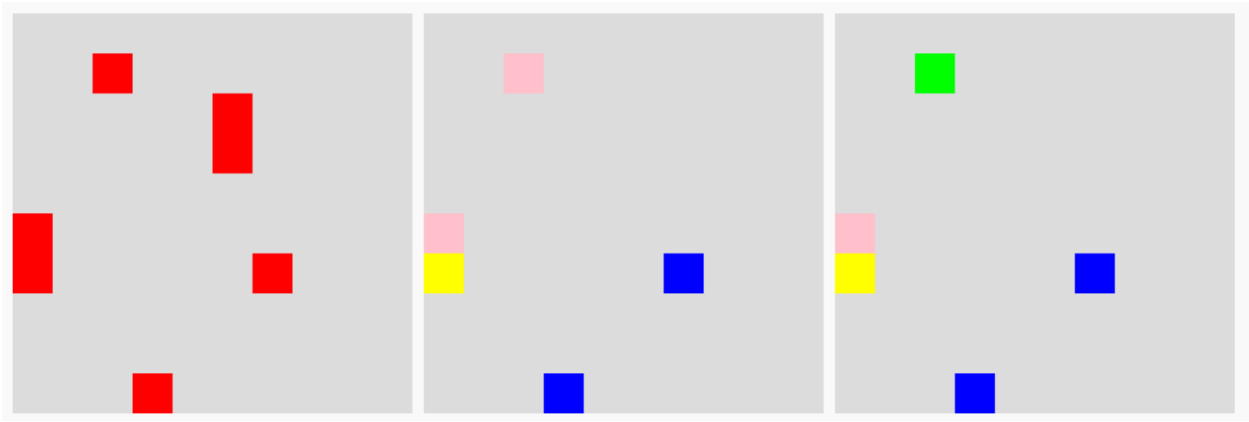
T=1 (Before first school formation)



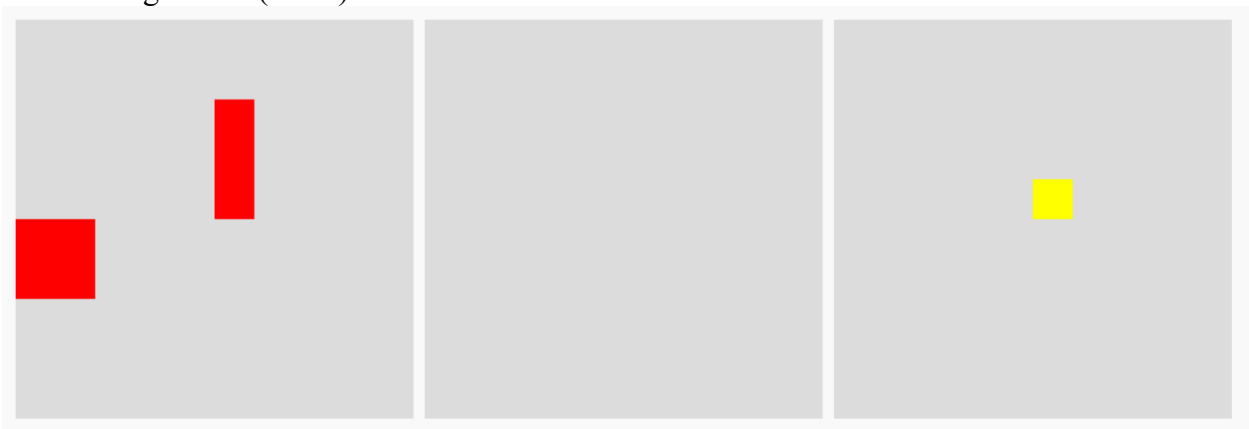
T=2



T=4



Final configuration (T=19)



Analysis & Discussion

We will briefly discuss each scenario below. Each is annotated with its Scenario number as found in the repository.

Vertical Alignment (Scenario 2)

This configuration sees three fish vertically aligned, with one cell separating each fish. This means that each fish is within each other's vision range.

Here we see that the most Northern fish acts as an anchor for the middle fish as they form a school at t=1. This confirms that movement North has priority over movement South

From here, we see that the Southern fish had initially moved North, but then moves South to avoid chasing the Northern school, which it has previously viewed as a single fish and not an actual school. This confirms the chase logic is performing as intended.

Horizontal Alignment (Scenario 3)

Here we see a similar configuration to Scenario 2, but in a horizontal orientation.

Here we see that the middle fish moves west to form a school with the fish to the West, confirming Western movement priority.

From here, we once again see anti-chase logic being proven as the Eastern fish sees its neighbor moving away from it and decides to go a different direction before later joining the school.

Cross Alignment (Scenario 4)

This configuration has the fish initially aligned in a cross formation, with an empty cell between each fish.

Here we see the middle fish is given priority to move West, forming a school with the fish to its West. The Eastern and Southern fish move West and North respectively. The Northern fish has been randomly assigned to move West, which is completes without competition.

From here, we see the Southern fish is randomly assigned to move West where it joins the school. The Eastern and Northern fish are both pulled into the school as well.

This situation showcases the Western priority as the center fish does not remain anchored and instead moves West.

Diamond Alignment (Scenario 5)

This configuration has the fish initially positioned into a diamond formation.

At t=1, we see once again that movement West is given priority as the Eastern fish collapses into the center of the diamond. This results in a three fish school. This is a great example of this priority as each fish has a claim to the center cell, but it is the fish moving West that is given priority over the others.

Right Angle Alignment (Scenario 6)

This configuration has the fish initially setup in a right-angle formation, with an empty cell between each fish.

This scenario showcases how the North Western fish ignores the other two fish in its field of vision, and instead acts as the anchor fish in the school formation.

Scattered Initialization (Scenario 1)

This is the most realistic scenario of the six, with all the fish scattered around the grid.

This scenario showcases how the random movement mechanic works, as each time a fish does not see a neighboring fish it moves in its assigned random direction. Additionally, this scenario showcases how all the rules work together in a very random, chaotic system as the final result is two distinct fish schools that form over time.

Future Extensions

As discussed in the original work, the addition of a predator could be introduced into the model. A reference for this behaviour can be found here [2].

References

[1] S. Hubbard, P. Babak, S. Th. Sigurdsson, and K. G. Magnússon, "A model of the formation of fish schools and migrations of fish," *Ecol. Model.*, vol. 174, no. 4, pp. 359–374, Jun. 2004, doi: 10.1016/j.ecolmodel.2003.06.006.

[2] M. Zheng, Y. Kashimori, O. Hoshino, K. Fujita, and T. Kambara, "Behavior pattern (innate action) of individuals in fish schools generating efficient collective evasion from predation," *J. Theor. Biol.*, vol. 235, no. 2, pp. 153–167, Jul. 2005, doi: 10.1016/j.jtbi.2004.12.025.